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Phase 3 Trial of Epicutaneous Immunotherapy with Viaskin Peanut in Toddlers

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Abstract:

Background: No approved peanut allergy treatment exists for children younger than 4 years. This study assessed efficacy and safety of epicutaneous immunotherapy (EPIT) with a peanut patch (Viaskin Peanut 250 µg [VP250]) among peanut-allergic toddlers.

Methods: This was a phase 3, multicenter, randomized, double-blind, placebo-controlled trial in peanut-allergic children aged 1–3 years,

confirmed by double-blind, placebo-controlled food challenge. Participants with eliciting dose (ED) ≤300 mg peanut protein were randomized 2:1 to 12 months of daily VP250 (n=244) or placebo (n=118). The primary endpoint was response, defined as Month 12 peanut challenge ED ≥1000 mg (if baseline ED >10 mg) or ≥300 mg (if baseline ED ≤10 mg). Safety was assessed by treatment-emergent adverse events (TEAEs).

Results: Of 362 participants randomized (median age: 2.5 years, 68.8% male, median ED 100 mg), 84.8% completed treatment. The primary efficacy endpoint was achieved by 67% vs 33.5% placebo participants (risk difference=33.4%; 95% CI: 22.4–44.5%, P<0.001). TEAEs, irrespective of relatedness, occurred in 100% of VP250 and 99.2% of placebo participants, including serious TEAEs in 8.6% of VP250 vs 2.5% of placebo participants and anaphylaxis in 7.8% of VP250 vs 3.4% of placebo participants. Serious treatment-related TEAEs occurred in 0.4% of VP250 vs no placebo participants, and treatment-related anaphylaxis in 1.6% of VP250 vs no placebo participants.

Conclusions: In this study of peanut-allergic children aged 1-3, 12 months of treatment with EPIT was superior to placebo in desensitizing children to peanuts, increasing the peanut dose triggering allergic symptoms.

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Phase 3 Trial of Epicutaneous Immunotherapy with Viaskin Peanut in Toddlers

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ABSTRACT

Background: No approved peanut allergy treatment exists for children younger than 4 years. This study assessed efficacy and safety of epicutaneous immunotherapy (EPIT) with a peanut patch (Viaskin Peanut 250 µg [VP250]) among peanut-allergic toddlers.

Methods: This was a phase 3, multicenter, randomized, double-blind, placebo-controlled trial in peanut-allergic children aged 1–3 years, confirmed by double-blind, placebo-controlled food challenge. Participants with eliciting dose (ED) ≤300 mg peanut protein were randomized 2:1 to 12 months of daily VP250 (n=244) or placebo (n=118). The primary endpoint was response, defined as Month 12 peanut challenge ED ≥1000 mg (if baseline ED >10 mg) or ≥300 mg (if baseline ED ≤10 mg). Safety was assessed by treatment-emergent adverse events (TEAEs).

Results: Of 362 participants randomized (median age: 2.5 years, 68.8% male, median ED 100 mg), 84.8% completed treatment. The primary efficacy endpoint was achieved by 67% vs 33.5% placebo participants (risk difference=33.4%; 95% CI: 22.4–44.5%, $P<0.001$). TEAEs, irrespective of relatedness, occurred in 100% of VP250 and 99.2% of placebo participants, including serious TEAEs in 8.6% of VP250 vs 2.5% of placebo participants and anaphylaxis in 7.8% of VP250 vs 3.4% of placebo participants. Serious treatment-related TEAEs occurred in 0.4% of VP250 vs no placebo participants, and treatment-related anaphylaxis in 1.6% of VP250 vs no placebo participants.

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Conclusions: In this study of peanut-allergic children aged 1-3, 12 months of treatment with EPIT was superior to placebo in desensitizing children to peanuts, increasing the peanut dose triggering allergic symptoms.

Trial Registration: clinicaltrials.gov: NCT03211247

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INTRODUCTION

Peanut allergy affects approximately 2% of children in the United States, Canada, and other westernized countries, with a rapidly rising prevalence over the past 20 years.[1-4] Typically lifelong,[5,6] it is a common cause of pediatric anaphylaxis (with potentially severe consequences, including fatality),[7-10] and has significant economic costs.[11,12] Peanut is a common culinary ingredient, complicating strict avoidance; severe reactions can occur from unintended exposure to small quantities.[13-15] Living with peanut allergy is often associated with adverse psychosocial effects in patients and caregivers.[16-18] Families have indicated treatments protecting their child against reactions from unintentional peanut exposures would improve quality of life and their child's socialization.[19]

While treatment strategies are emerging, there are no approved therapies for children under 4 years of age. Desensitization, defined as an increase in reaction dose threshold to a food allergen (thereby reducing reaction risk from unintended ingestions), is a primary treatment goal.[20,21] Early peanut consumption during infancy can substantially reduce the risk of developing peanut allergy.[22] Thus, at these ages, the developing immune system may be particularly amenable to desensitization,[23-26] providing important rationale for prioritizing treatments targeting younger children. Peanut oral immunotherapy (OIT) is available in non-commercialized forms for children <4 years and can induce desensitization,[23,24] but is associated with potentially arduous dosing schedules and risk of treatment-associated systemic adverse events (AEs) such as anaphylaxis and gastrointestinal events,[27-29] creating a need for approaches with alternative routes of administration and different safety profiles.

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Epicutaneous immunotherapy (EPIT) with Viaskin Peanut 250 µg (VP250), a patch-based non-oral immunotherapy, has consistent evidence of efficacy, safety, and high treatment adherence in peanut-allergic children aged 4–11 years.[30-32] EPIT accomplishes this with substantially lower allergen doses (microgram vs milligram) vs other therapeutic alternatives. There is a need for safe, effective desensitization treatment with a low adverse-event profile for very young peanut-allergic children. We aimed to assess the efficacy and safety of EPIT with VP250 among peanut-allergic children aged 1–3 years in the phase 3 trial EPIT in Toddlers with Peanut Allergy (EPITOPE).

METHODS

Trial Design and Participants

This was a multicenter, double-blind, placebo-controlled, randomized phase 3 trial to assess safety and efficacy of 12 months of daily EPIT with VP250 in peanut-allergic children aged 1-3 years (Figure S1). VP250 safety and dose optimization in 1–3-year-old children was established in a prior sub-study of 51 peanut-allergic children randomized to receive 12 months of placebo, 100 µg or 250 µg peanut protein-containing Viaskin Peanut, with the 250-µg dose approved by an independent data safety monitoring board.[33] DBV Technologies designed (with investigator input) and sponsored the trial, with data gathered by site principal investigators (who are authors) and analyzed by members of the writing committee (roles in Supplementary Appendix); all authors participated in the decision to publish, contributed to writing/editing, and vouch for the data. Drs. Greenhawt and Campbell wrote the first draft. The protocol stipulated investigators could not independently publish results without consultation with the sponsor.

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The protocol and informed consent were approved by institutional review boards; the protocol and statistical analysis plan are available with this article at nejm.org. Written informed consent was provided by a legally authorized representative. The study was conducted between July 31, 2017, and April 27, 2022, in 8 countries (51 sites) across the United States, Canada, Australia, and Europe.

Key inclusion criteria were: (i) physician-diagnosed peanut allergy; (ii) peanut-specific immunoglobulin E (sIgE) >0.7 kU/L and skin prick test (largest wheal diameter) ≥6 mm; (iii) peanut protein eliciting dose of ≤300 mg at double-blind, placebo-controlled food challenge. Complete inclusion/exclusion criteria are outlined in Table S1. Screening and Month 12 (M12) peanut double-blind placebo-controlled food challenge were conducted per PRACTALL guidelines, using a standardized, blinded food matrix.[34] At screening peanut challenge, peanut protein doses of 1, 3, 10, 30, 100, and 300 mg (maximum dose) were administered every 30 minutes; the challenge was stopped when signs/symptoms met prespecified stopping criteria (Table S2). The dose triggering symptoms meeting stopping rules was considered the eliciting dose. For M12 food challenge, 1000-mg and 2000-mg doses were added (maximum cumulative dose=3444 mg). Randomization was stratified by site. Treatment adherence was defined as the total number of patches applied in the treatment period divided by the number of days in that period.

Interventions

VP250 is an epicutaneous system (patch) consisting of unmodified lyophilized peanut protein within an occlusive chamber (Figure S2A). Participants were randomized 2:1 to VP250 or placebo patch (identical in size, shape, appearance, differing only in peanut protein presence)

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applied daily to the interscapular area (Figure S2B) for 12 months. The first patch was applied for 3 hours under medical supervision. Daily application duration was titrated at home over 4 weeks (Titration Period) according to the schedule in Figure S2. No restrictions on any activities, including bathing or swimming, were required while wearing the patch.

Outcomes

The primary measure of treatment effect compared responder percentages between the VP250 and placebo groups after 12 months of EPIT treatment. A subject was defined as a treatment responder if the baseline eliciting dose was >10 mg peanut protein and the post-treatment eliciting dose was ≥1000 mg peanut protein (~3-4 peanuts), or if the baseline eliciting dose was ≤10 mg peanut protein and the post-treatment eliciting dose was ≥300 mg peanut protein (~1 peanut). The two key secondary endpoints included in a hierarchical plan to adjust for multiple testing were change from baseline to M12 in cumulative reactive dose and eliciting dose in the VP250 vs placebo group. Other prespecified secondary endpoints were: (i) response defined by M12 eliciting dose ≥1000 mg peanut protein, regardless of the screening eliciting dose; (ii) response defined by M12 cumulative reactive dose ≥3444 mg peanut protein in each group; and (iii) change in severity of symptoms elicited during the challenge from baseline to M12 in each group. Treatment-emergent adverse events (TEAEs) were evaluated for all participants receiving at least 1 dose of study drug (safety set). TEAEs were graded mild, moderate, or severe and serious according to Common Terminology Criteria for Adverse Events.[35] TEAEs were classified according to the Medical Dictionary for Regulatory Activities (MedDRA) V24. Local application-site reactions were graded according to a 5-point scale (0–4) per protocol (Section 6.2.6). Adverse events of special interest (AESIs) were defined as local (patch-site reactions,

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potentially leading to skin barrier disruption or Grade 4 reactions) or systemic (any acute systemic allergic reaction not occurring on food challenge, regardless of causal relationship to patch). Anaphylactic reactions were defined by NIAID criteria[36] and/or by investigator report of "anaphylaxis" or "anaphylactic reaction", regardless of severity and causal relationship to VP250.

Statistical Power and Sample Size

A sample of 350 subjects (n=233 VP250, n=117 placebo) was estimated to achieve 90% power to detect a lower bound (LB) of the 95% confidence interval (CI) of at least a 15% absolute risk difference using the Newcombe method for CI computation. A 40% VP250 response rate and 10% placebo response rate (30% absolute risk difference) was assumed for comparing the primary responder endpoint after 12 months of treatment.

Statistical Methods

The primary analysis included all participants according to their randomized assignment (full analysis set) and was assessed using a 2-sided Farrington-Manning 95% CI for the difference in response rates between the randomized groups. For the primary analysis, participants with missing M12 food challenge values were imputed according to the type of intercurrent event (Table S3). If statistical significance was observed for the primary outcome, to control the family-wise error rate at 5%, a hierarchical testing procedure was applied to testing two secondary endpoints (peanut cumulative reactive dose at M12, peanut eliciting dose at M12). Other efficacy and subgroup analyses that were not subject to multiple testing corrections have been presented as point estimates with 95% CI. The widths of the intervals were not adjusted for

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multiplicity and should not be used in place of hypothesis testing. Prespecified sensitivity analyses were performed as described in the statistical analysis plan. All analyses presented were prespecified in the statistical analysis plan, except Figure S3, which is descriptive and post-hoc.

RESULTS

Study Participants, Treatment Exposure, and Adherence

The population for primary efficacy and safety analysis consisted of all 362 randomized participants (68.8% male; median age [Q1, Q3]: 2.5 years; [1.7, 3.2], 63.3% White race), with 244 and 118 participants randomized to VP250 and placebo groups, respectively (Figure 1) and analyzed per treatment assignment. There were 67 participants in the eliciting dose subgroup ≤10 mg and 295 participants in the eliciting dose subgroup >10 mg. Baseline demographic characteristics for the treatment and placebo groups were balanced (Table 1).

The overall treatment exposure was (median [Q1, Q3]) 376 days (368, 419). Post-titration period, the median [IQR] average daily application duration was 22.2 hours [19.4, 23.4] in the VP250 group and 23.7 hours [22.9, 24.0] in the placebo group. Treatment adherence was high and comparable between groups, with mean (standard deviation) adherence rate of 97.0% (6.9); 84.8% of participants (208/244 VP250-randomized and 99/118 placebo-randomized) completed study treatment.

Assessment of Clinical Response

<u>Primary endpoint</u>

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At M12, among all randomized participants, a significantly larger percentage of VP250 participants achieved the primary endpoint (67%) vs placebo (33.5%), with a difference of 33.4% (95% CI, 22.4–44.5; *P*<0.001) (Figure 2A). In a post-hoc analysis, a treatment effect was suggested across all baseline eliciting doses >1 mg (Figure S3). All prespecified primary endpoint sensitivity analyses were also consistent with the primary analysis (Figure 2B), including those participants with treatment and assessment peanut challenges outside the M12 window (primarily attributable to COVID-19–related study site access). The results of all prespecified subgroup analyses of the primary endpoint are shown in Figure S4.

Secondary and Exploratory Endpoints

Two key secondary endpoints were assessed hierarchically. VP250 recipients achieved a median [Q1, Q3] change in cumulative reactive dose from baseline to M12 of 1300 mg [140, 3000] vs 0 mg [0, 1000] for placebo recipients. By ANCOVA, the geometric least squares (LS) means ratio was 3.13 (*P*<0.001). Secondly, VP250 recipients achieved a median [Q1, Q3] change in eliciting dose from baseline to M12 of 900 mg [90, 1700] vs 0 mg [0, 700] for placebo recipients. By ANCOVA, the geometric LS means ratio was 2.96 (*P*<0.001).

In addition, 64.2% VP250 vs 29.6% placebo participants (difference 34.7%, 95% CI, 23.6–45.7) had a M12 eliciting dose of ≥1000 mg regardless of their baseline eliciting dose subgroup; 37% VP250 vs 10.0% placebo participants reached a M12 cumulative reactive dose ≥3444 mg (difference 27.0%, 95% CI, 17.9–36.1). At baseline challenge, proportions of mild/moderate/severe objective signs/symptoms between VP250 and placebo participants were balanced. At M12, the distribution of maximum symptom severity was shifted to less severe

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symptoms in VP250 vs placebo, coincident with an increased eliciting dose and more responders in the active group vs placebo (Figure 3, Figure S5).

Peanut-sIgE decreased from baseline to M12 in VP250 participants and increased in placebo participants. In contrast, peanut-sIgG4 increased from baseline to M12 in VP250 participants with little change in placebo participants (Figure S6).

Safety Assessments

TEAEs occurred in 100% of VP250 participants and 99.2% of placebo participants (Table 2, Table S4). The most common TEAEs were application-site reactions, including erythema (VP250: 98.0%; placebo: 90.7%); pruritus (VP250: 94.7%; placebo: 61.0%), and site swelling (VP250: 72.5%; placebo: 39.0%). As per investigator assessment, skin reactions were more frequent at the beginning of treatment (Months 0–3) but then decreased in frequency and were mostly Grade 1 (erythema, or erythema and infiltration) or 2 (erythema and few papules); one VP250 participant (0.4%) had a local AESI Grade 4 (erythema, vesicles) at Month 3 visit. Local TEAEs lasting >90 days occurred in 81.6% of VP250 vs 55.1% of placebo participants. Topical corticosteroids for treatment of local application-site reactions were used at least once during the treatment period in 95.9% VP250 and 63.6% placebo participants (Table 2, Table S5).

Twenty-one (8.6%) VP250 vs 3 (2.5%) placebo participants reported serious AEs (SAE) (excluding during food challenge) (Table S6). One (0.4%) VP250 participant reported an SAE (periorbital edema) deemed treatment-related.

Site investigators considered 26 anaphylactic reactions (occurring outside food challenges) as systemic AESI in 20 participants, irrespective of the assessed causal relationship (VP250: 17 [7%]; placebo: 3 [2.5%]; summaries are presented in Table S7A, some of which included allergic

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reactions to non-peanut food allergens). Four (1.6%) were deemed treatment-related by investigators, all occurring in VP250 participants; 3 were treated with a single dose of epinephrine, one with no epinephrine, and all were of mild or moderate severity. In 2 participants (both age 3), this occurred on the first day of patch application; one received epinephrine and discontinued study participation. Both had 100 mg eliciting doses and peanut sIgE levels above the study population median (Table S7B). No anaphylactic reactions occurred during sleep.

Accidental peanut consumption occurred in 24 (9.8%) VP250 participants and 15 (12.7%) placebo participants. Rates of allergic reactions and anaphylaxis due to accidental peanut consumption (by investigator report) were lower for VP250 vs placebo: 13 (5.3%) vs 12 (10.2%), and 3 (1.2%) vs 3 (2.5%), respectively.

Fourteen TEAEs led to permanent study discontinuation (8 [3.3%] VP250 and 0 placebo participants). In 7 participants, these TEAEs were deemed possibly related or related, including local application-site reactions (n=4), eczema (n=1), anaphylaxis (n=1), and abnormal behavior/application-site reaction/eczema (n=1).

DISCUSSION

In this phase 3 study, VP250 treatment vs placebo was superior in desensitizing peanut-allergic children aged 1–3 years, with consistent efficacy findings across numerous subgroup and sensitivity analyses. Anaphylaxis was experienced by 19 children in the VP250 group and 4 in the placebo group, and was judged by investigators as treatment-related in no placebo participants and 4 VP250 participants, 1 of whom discontinued treatment; the other 3 resumed treatment without further anaphylaxis episodes.

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This trial demonstrated peanut desensitization using a non-oral immunotherapy. A previously published phase 3 trial in children aged 4–11 years demonstrated that EPIT with VP250 was associated with a low occurrence of anaphylaxis.[30-32] Consistent with prior VP250 studies, TEAEs in EPITOPE were experienced by most participants, but were primarily mild-to-moderate local skin reactions, decreasing in frequency with study duration. Although severe local application-site skin reactions occurred in more VP250 participants vs placebo, these infrequently led to study discontinuation, and improved over time.

There are few other published studies of peanut allergy treatments in toddlers and young children, including a blinded and placebo-controlled trial.[23,24] A recently published multi-center double-blinded, placebo-controlled peanut flour OIT trial in children aged 1–4 years reported a significant treatment effect, with treatment-related epinephrine use of 22% and 3% rate of eosinophilic esophagitis.[23] Similar to reports in older children, AEs in that OIT trial were primarily oral, gastrointestinal, lower respiratory, and systemic.[37-39]

While EPITOPE has a high rate of placebo responders, this study was conducted in children at an age where natural resolution is expected to be highest and is consistent with recent Australian population-level data noting 29% of children outgrow peanut allergy by age 6 years.[6] Moreover, the placebo rate includes those participants who outgrew their allergy, but also those who (moving toward natural resolution) increased their threshold but remained allergic to higher peanut doses, as suggested by a 10% placebo rate in children who tolerated ≥3444 mg of peanut protein at M12 challenge.

LIMITATIONS

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This study has several limitations. First, participants with a history of severe peanut anaphylaxis (eg, intubation and ventilation, inotropic support) were excluded, a similar safety exclusion across other food allergy treatment trials employing food challenge. This could limit generalizability, although real-world study of VP250 in 4–11-year-old children, which included patients with severe anaphylaxis, noted no major safety concerns.[32] Second, consistent with other food allergy treatment studies, a lack of racial diversity among participants may limit generalizability. While more males than females were randomized, this is consistent with other trials and reflects atopy rates in childhood.[12] A summary of representativeness of patient sample is presented in Table S8. Third, the optimal duration of VP250 use for maximal treatment response is unknown, and sustained effects following treatment cessation were not assessed. Data from the open-label extension in the 4–11-year-old pivotal trial showed desensitization effects are enhanced with longer treatment.[31] A 24-month, open-label extension study of EPITOPE to determine continued treatment response is ongoing (NCT03859700), which has enrolled 88% of all eligible EPITOPE participants.

CONCLUSION

In this phase 3 study of peanut-allergic children aged 1–3 years, 12 months of daily EPIT with a 250-µg peanut patch (approximately 1/1000 of one peanut) resulted in significantly more children achieving desensitization vs placebo, sufficient to lessen the likelihood of allergic reactions upon accidental exposure.

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Data Sharing Statement: A data sharing statement provided by the authors is available with this article at nejm.org.

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TABLES AND FIGURE LEGENDS

Table 1. Demographics and Baseline Characteristics.

Category	VP250 (N=244)	Placebo (N=118)
Age, years, median (Q1, Q3)	2.50 (1.75, 3.20)	2.40 (1.70, 3.10)
Gender, n (%)		
Male	165 (67.6)	84 (71.2)
Female	79 (32.4)	34 (28.8)
Race/ethnic origin, n (%)		
White	159 (65.2)	70 (59.3)
Asian	41 (16.8)	24 (20.3)
Black or African American	1 (0.4)	1 (0.8)
Native Hawaiian or other Pacific Islander	2 (0.8)	0
Other	33 (13.5)	17 (14.4)
Missing	8 (3.3)	6 (5.1)
Peanut-specific IgE, kU_A/L		
Median (Q1, Q3)	13.4 (4.04, 65.85)	14.75 (4.86, 52.11)
Range	0.8–971.0	0.7–1031.0
Peanut-specific IgG4, mg/L		
Median (Q1, Q3)	0.38 (0.13, 1.29)	0.41 (0.14, 1.04)
Range	0.1–76.7	0.1–16.3
Peanut protein eliciting dose, mg		
Median (Q1, Q3)	100 (30, 300)	100 (30, 300)
Range	1–300	1–300
Baseline eliciting dose, n (%)		
1 mg	7 (2.9)	1 (0.8)
3 mg	17 (7.0)	3 (2.5)
10 mg	27 (11.1)	12 (10.2)
30 mg	27 (11.1)	17 (14.4)
100 mg	72 (29.5)	37 (31.4)
300 mg	94 (38.5)	48 (40.7)
Medical history, n (%)		
Asthma	39 (16.0)	27 (22.9)
Eczema/atopic dermatitis	194 (79.5)	96 (81.4)
Allergic rhinitis	49 (20.1)	23 (19.5)
Food allergy(ies) other than peanut	161 (66.0)	81 (68.6)

IgE, immunoglobulin E; IgG4, immunoglobulin G4; kU_A/L, kilounits of antibody per liter; Q1, quartile 1; Q3, quartile 3.

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Table 2. Summary of Treatment-Emergent Adverse Events by Treatment Group (Safety Population).

	VP250 (N=244)		Placebo (N=118)	
TEAEs	**n**	**(%)**	**n**	**(%)**
Any TEAE	244	100	117	99.2
Related to IMP	244	100	112	94.9
Serious* TEAE	21	8.6	3	2.5
Related to IMP	1	0.4	0	0
Severe* TEAE	63	25.8	12	10.2
Related to IMP	57	23.4	11	9.3
Moderate TEAE	225	92.2	91	77.1
Related to IMP	208	85.2	59	50.0
Mild TEAE	242	99.2	117	99.2
Related to IMP	238	97.5	110	93.2
Any treatment-emergent local AESI†	48	19.7	12	10.2
Any treatment-emergent systemic AESI	22	9.0	5	4.2
System organ class‡				
Preferred term (select terms of interest)				
Administration site conditions	243	99.6	111	94.1
General disorders	93	38.1	52	44.1
Skin and subcutaneous disorders	163	66.8	82	69.5
Immune system disorders	86	35.2	25	21.2
Anaphylactic reaction	19	7.8	4	3.4

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Related to IMP	4	1.6	0	0
Non-anaphylactic hypersensitivity reaction	24	9.8	11	9.3
Related to IMP	3	1.2	0	0
Eye disorders	26	10.7	14	11.9
Infections and infestations	203	83.2	94	79.7
Gastrointestinal disorders	101	41.4	47	39.8
Respiratory, thoracic, and mediastinal disorders	123	50.4	58	49.2
Injury, poisoning, and procedural complications	63	25.8	33	28.0
TEAEs leading to temporary discontinuation	63	25.8	16	13.6
TEAEs leading to permanent discontinuation	8	3.3	0	0
TEAEs leading to epinephrine intake	25	10.2	8	6.8
TEAEs leading to systemic or inhaled corticosteroid use	69	28.3	35	29.7
TEAEs leading to topical corticosteroid use	234	95.9	75	63.6

AESI, adverse event of special interest; IMP, investigational medicinal product; n, number of participants experiencing at least 1 event; N, number of participants in treatment group; TEAE, treatment-emergent adverse event.

*TEAEs were graded mild, moderate, or severe and serious according to Common Terminology Criteria for Adverse Events.

†Local AESIs were defined as reactions at patch sites potentially leading to skin barrier disruption, and systemic AESIs were defined as: 1. anaphylaxis, and 2. systemic hypersensitivity reactions leading to epinephrine intake whatever the causal relationship to IMP.

‡System organ class classification according to Medical Dictionary for Regulatory Activities (MedDRA) V24. The table includes only TEAE reported to occur with 5% or greater frequency in either group.

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Figure Legends

Figure 1. Participant Disposition CONSORT Diagram.

Eligible participants were randomized 2:1 to receive Viaskin Peanut 250 µg or placebo.

VP250=Viaskin Peanut 250 µg.

Figure 2. Primary (A) and Sensitivity (B) Analyses of the Primary Endpoint.

A. The primary measure of treatment effect compared the percentages of treatment responders between the Viaskin Peanut 250 µg and placebo groups after 12 months of EPIT treatment among all randomized subjects. A subject was defined as a treatment responder if the baseline eliciting dose was >10 mg peanut protein and the Month 12 eliciting dose was ≥1000 mg peanut protein, or the initial eliciting dose was ≤10 mg peanut protein and the Month 12 eliciting dose was ≥300 mg peanut protein.

B. Prespecified sensitivity analyses of the primary endpoint were defined as 1: Primary efficacy analysis using missing=failure imputation method for intercurrent events 1, 2, and 5; 2: Primary efficacy analysis using multiple imputation method for intercurrent events 1, 2, and 5; 3: Primary efficacy analysis using multiple imputation method for intercurrent events 2 and 5 (active subjects mimic Viaskin Peanut 250 µg response and placebo subjects mimic placebo response); 4: Primary efficacy analysis excluding subjects with intercurrent event 1, 2, or 5; 5: Primary efficacy analysis excluding subjects whose double-blind, placebo-controlled food challenge falls outside the Month 12 time-window (intercurrent event 4); 6: Primary efficacy analysis on the per-protocol set.

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Figure 3. Maximum Symptom Severity at Double-Blind Placebo-Controlled Food Challenges.

The severity of symptoms elicited during the double-blind, placebo-controlled food challenge was measured at baseline and Month 12 in each treatment group. Severity was assessed by maximum grade of PRACTALL in any PRACTALL symptom domain at each dose.[1] Subjects with intercurrent events 1, 2, 3 and 5 were excluded from this analysis since a Month 12 peanut food challenge was not completed. Subjects who completed both the baseline and Month 12 double-blind, placebo-controlled food challenge were included in this analysis (200/244 randomized to VP250 and 98/118 randomized to placebo)

Reference

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Figure 1. Participant Disposition CONSORT Diagram.

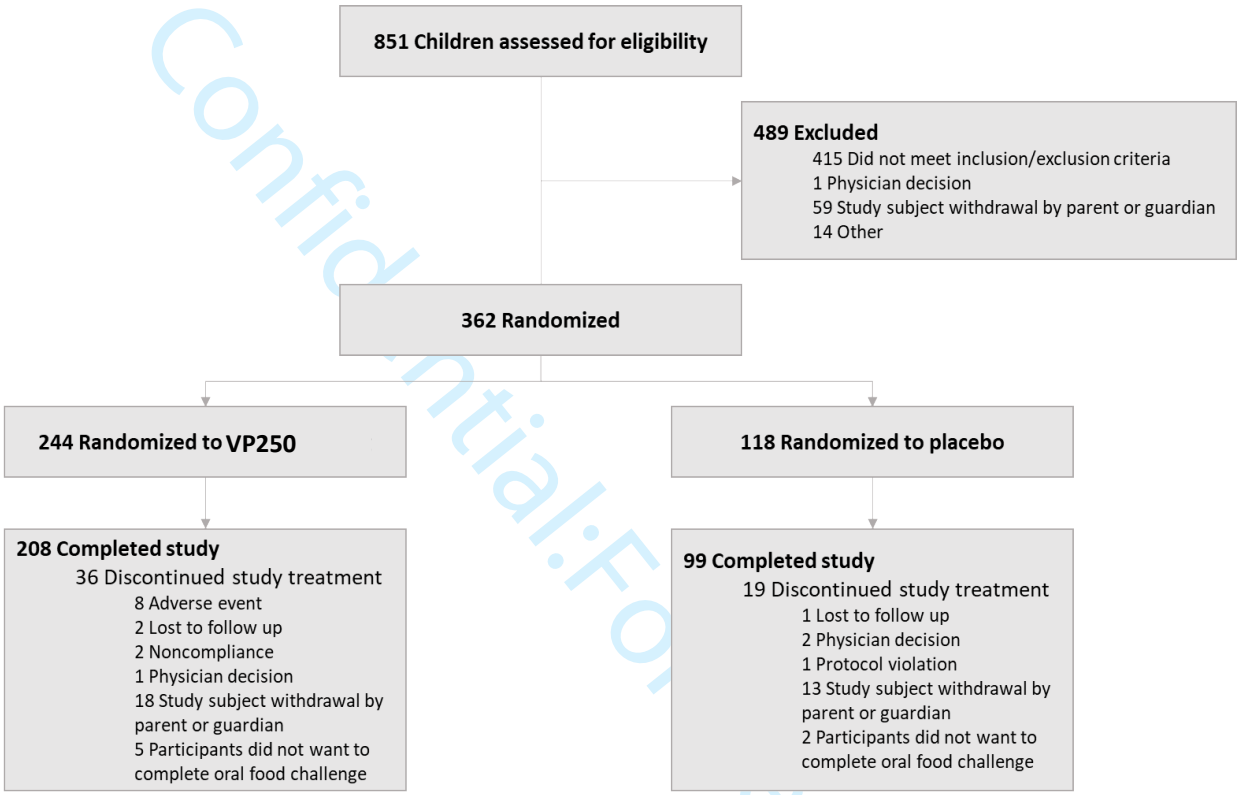

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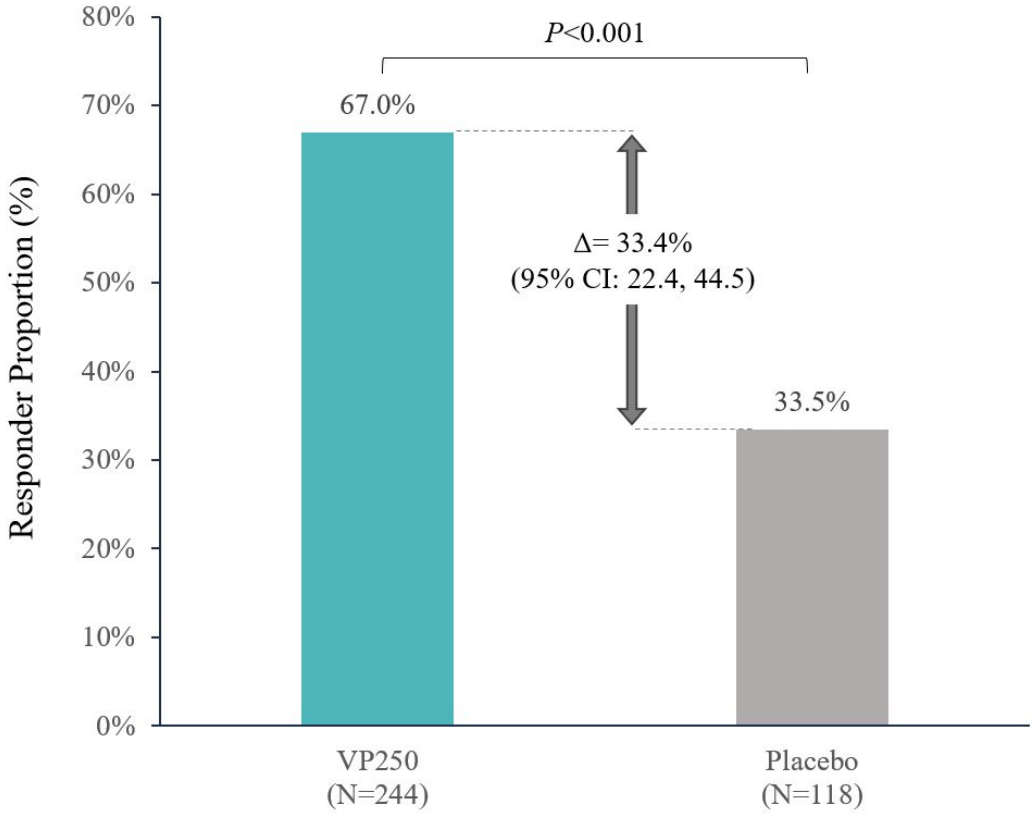
Figure 2. Primary (A) and Sensitivity (B) Analyses of the Primary Endpoint.

A.



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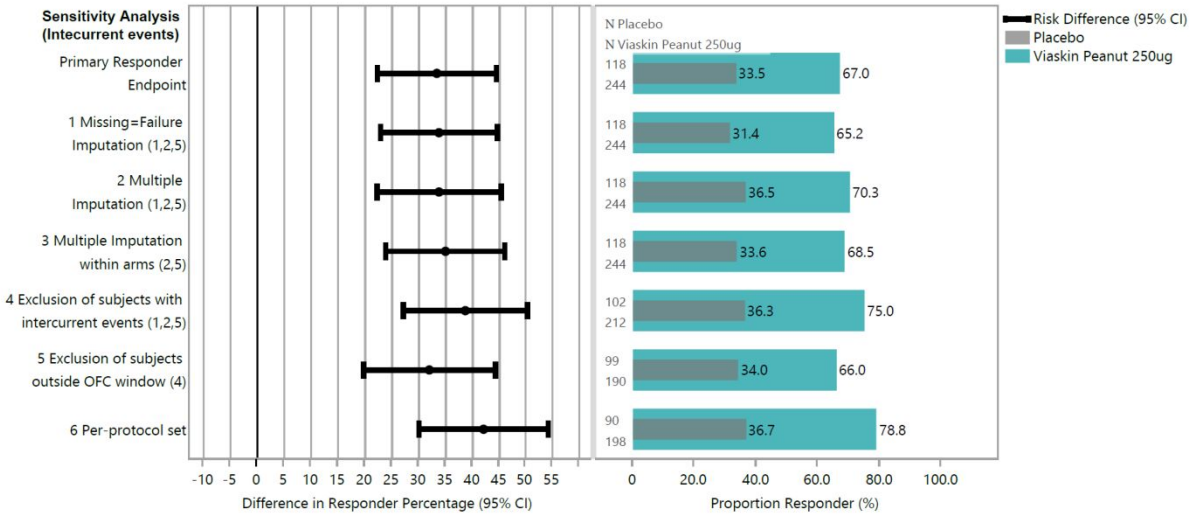

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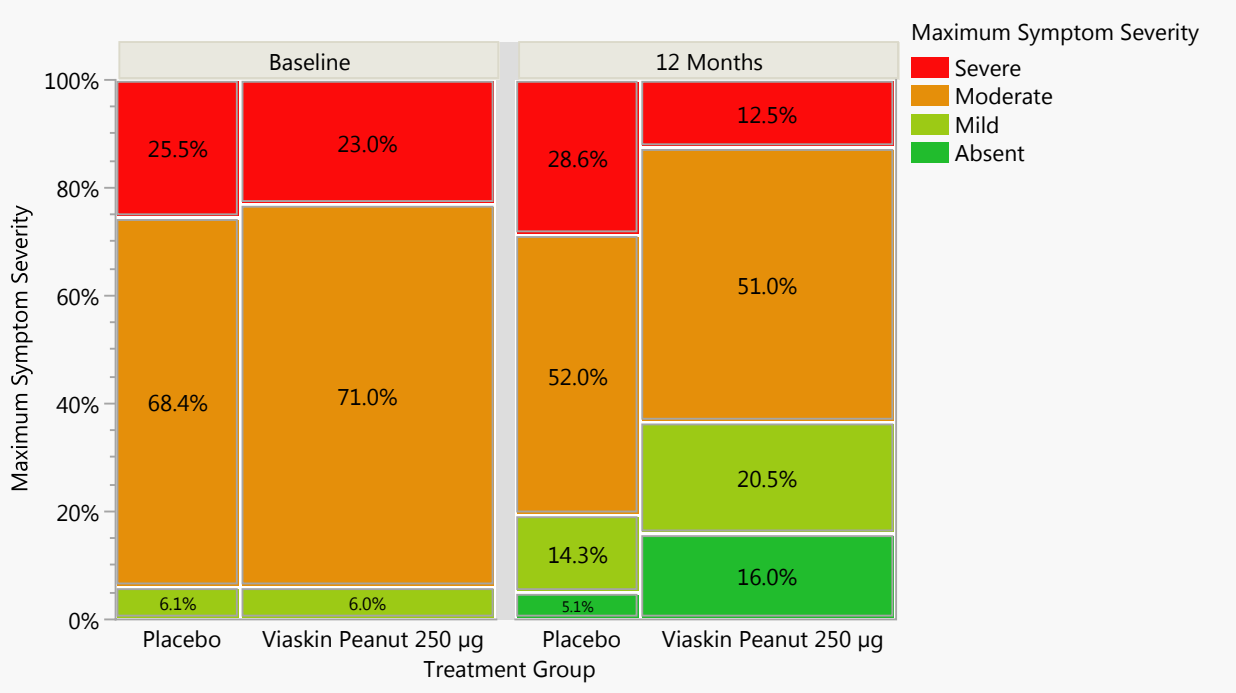
Figure 3. Maximum Symptom Severity at Month 12 Double-Blind Placebo-Controlled Food Challenge.



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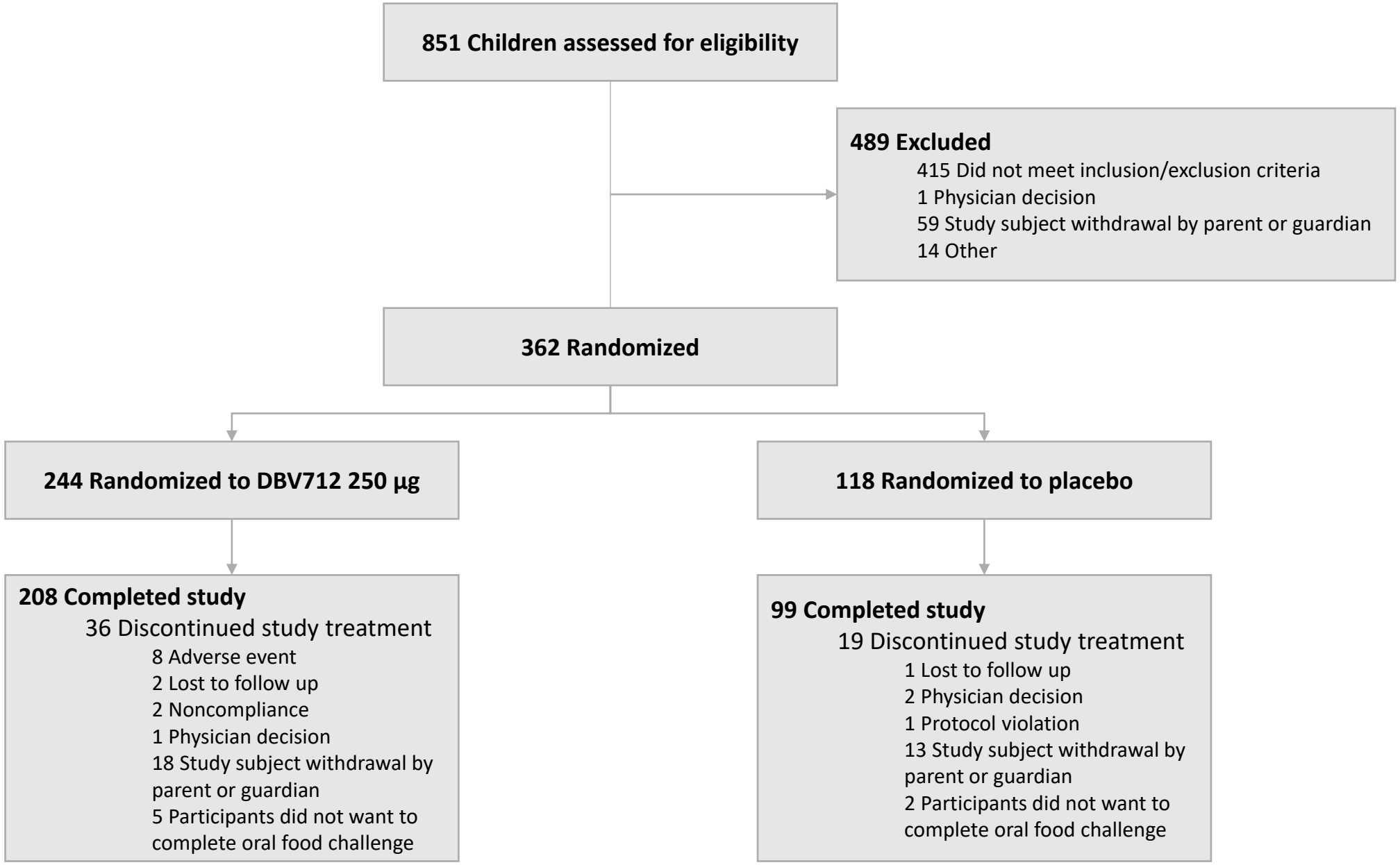
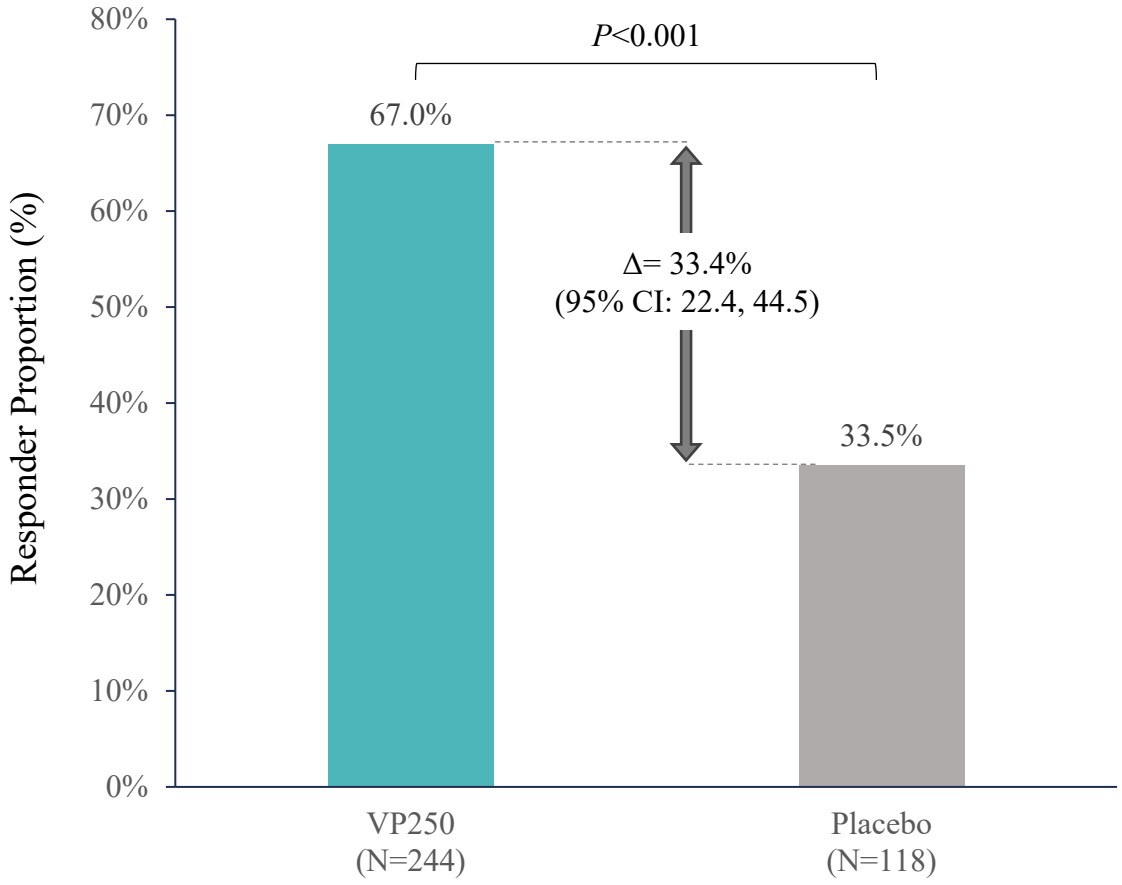
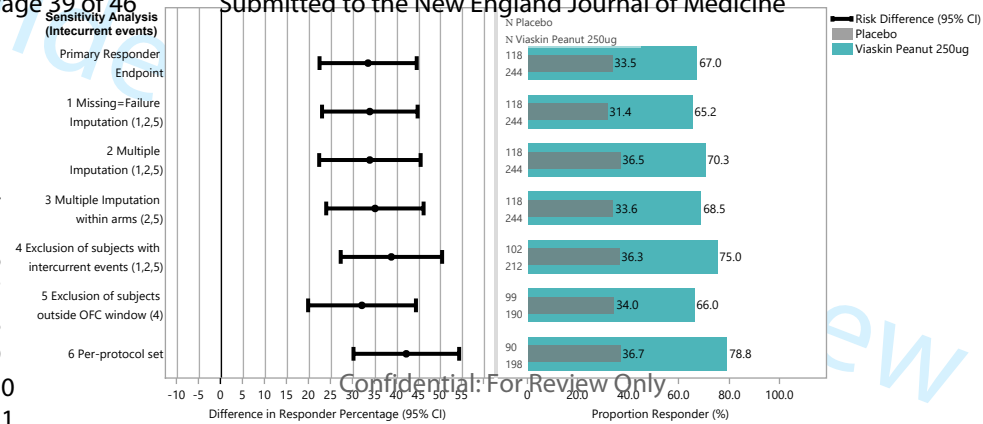

851 Children assessed for eligibility
489 Excluded
415 Did not meet inclusion/exclusion criteria
1 Physician decision
59 Study subject withdrawal by parent or guardian
14 Other
362 Randomized
244 Randomized to DBV712 250 µg
118 Randomized to placebo
208 Completed study
36 Discontinued study treatment
8 Adverse event
2 Lost to follow up
2 Noncompliance
1 Physician decision
18 Study subject withdrawal by parent or guardian
5 Participants did not want to complete oral food challenge
99 Completed study
19 Discontinued study treatment
1 Lost to follow up
2 Physician decision
1 Protocol violation
13 Study subject withdrawal by parent or guardian
2 Participants did not want to complete oral food challenge


80%
70%
60%
50%
40%
30%
20%
10%
0%
Responder Proportion (%)
P<0.001
67.0%
Δ= 33.4%
(95% CI: 22.4, 44.5)
33.5%
VP250
(N=244)
Placebo
(N=118)



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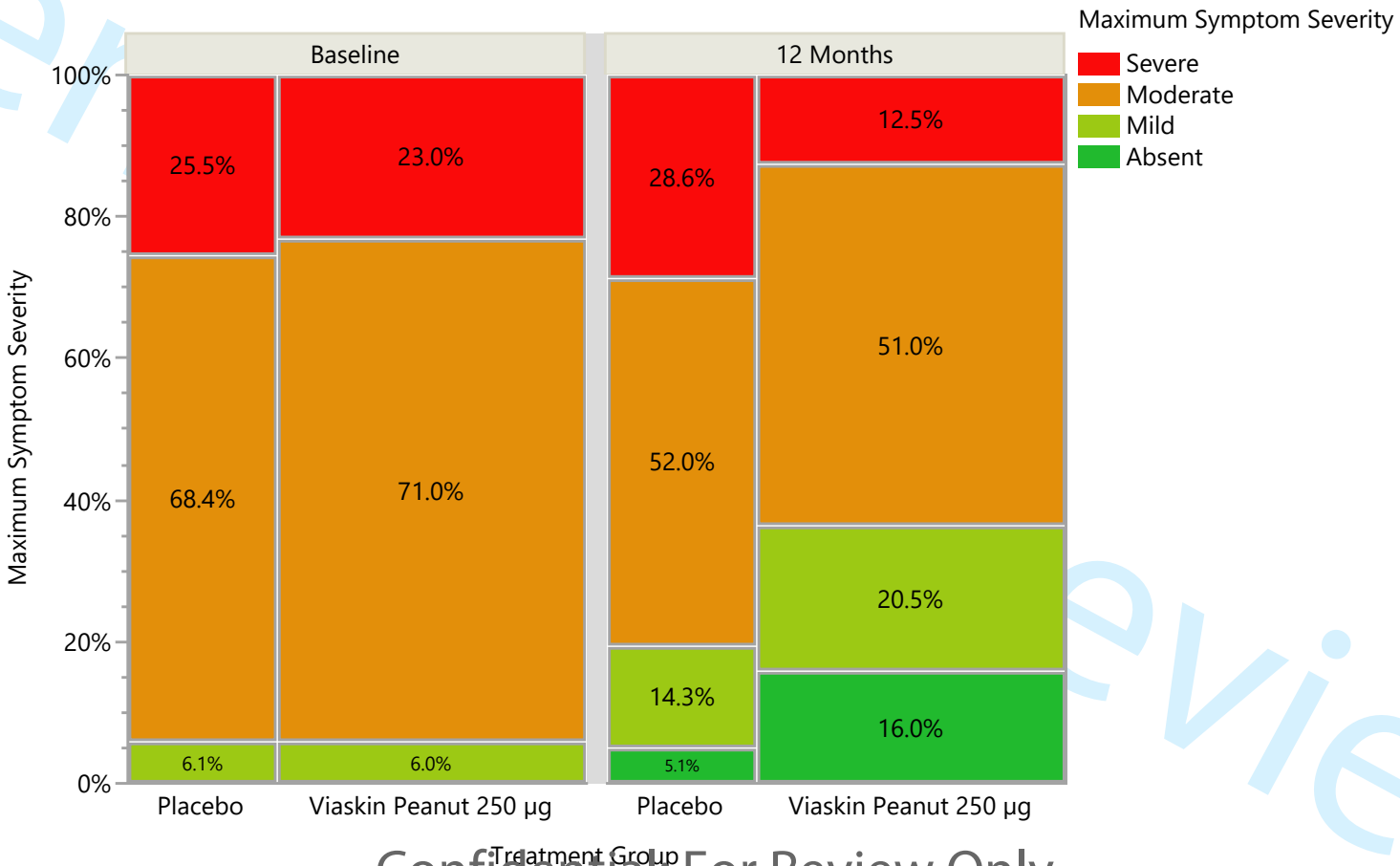

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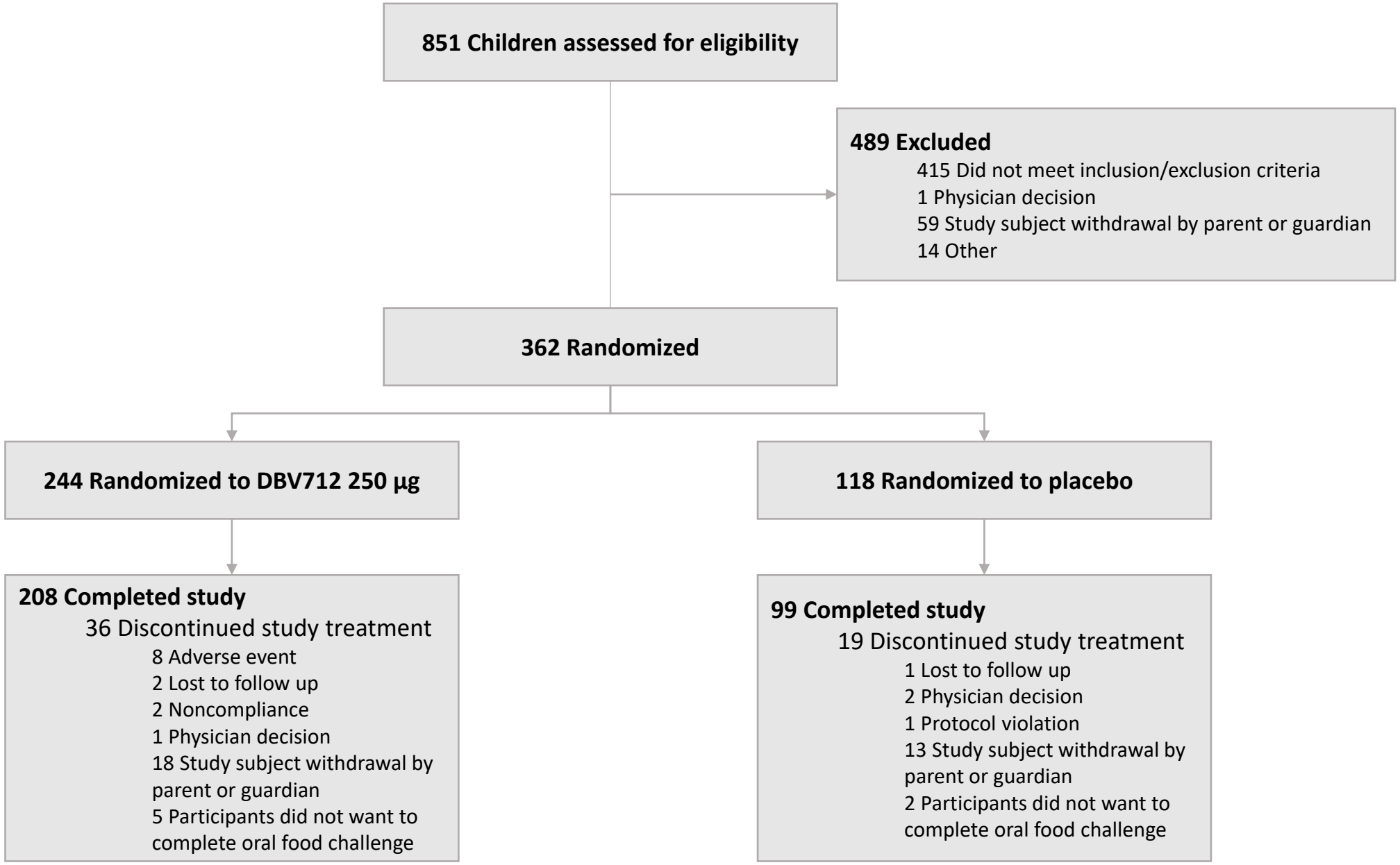
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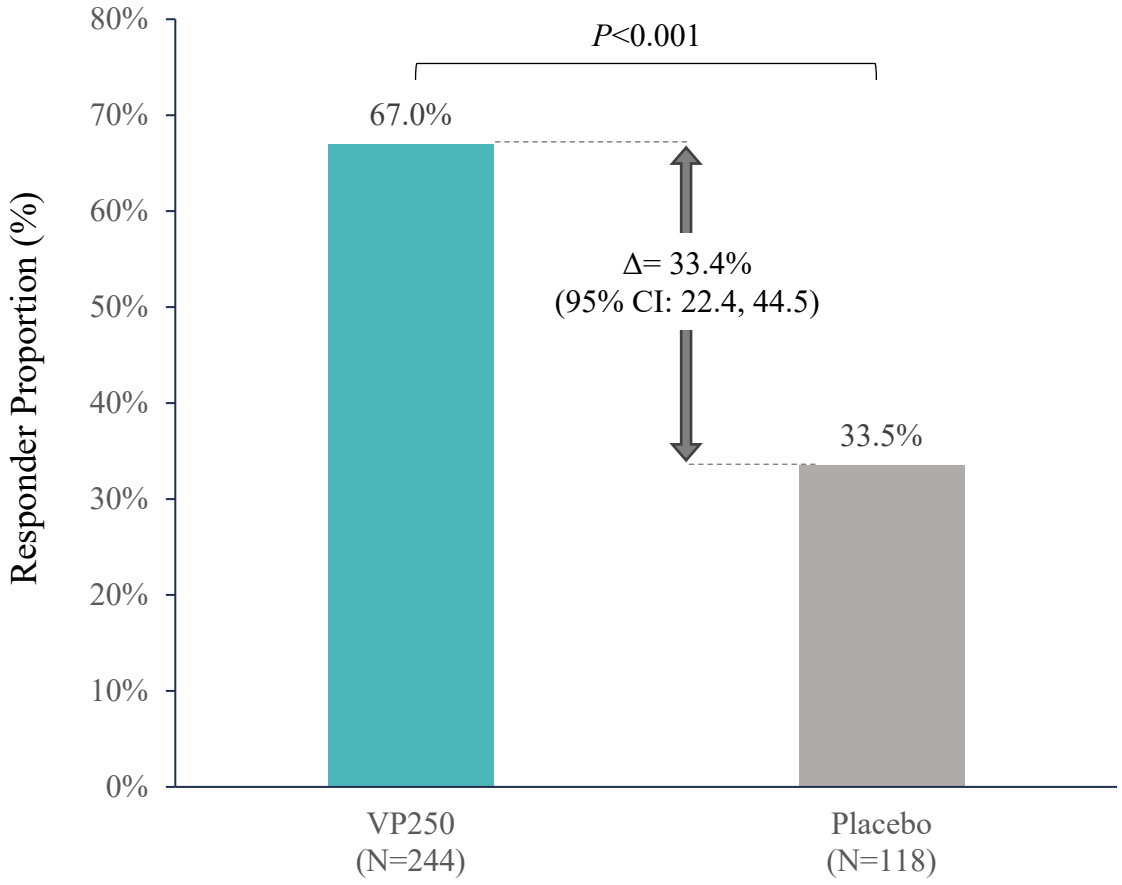
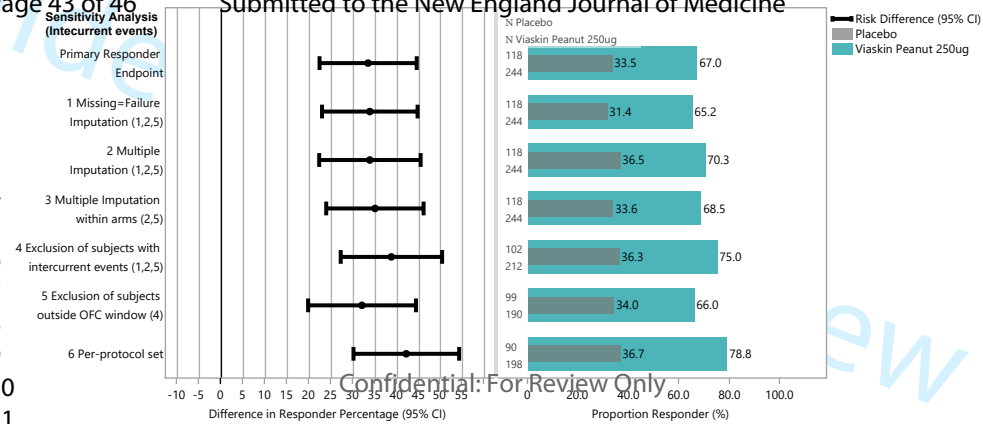
13 Study subject withdrawal by parent or guardian
2 Participants did not want to complete oral food challenge


80%
70%
60%
50%
40%
30%
20%
10%
0%
Responder Proportion (%)
P<0.001
67.0%
Δ= 33.4%
(95% CI: 22.4, 44.5)
33.5%
VP250
(N=244)
Placebo
(N=118)



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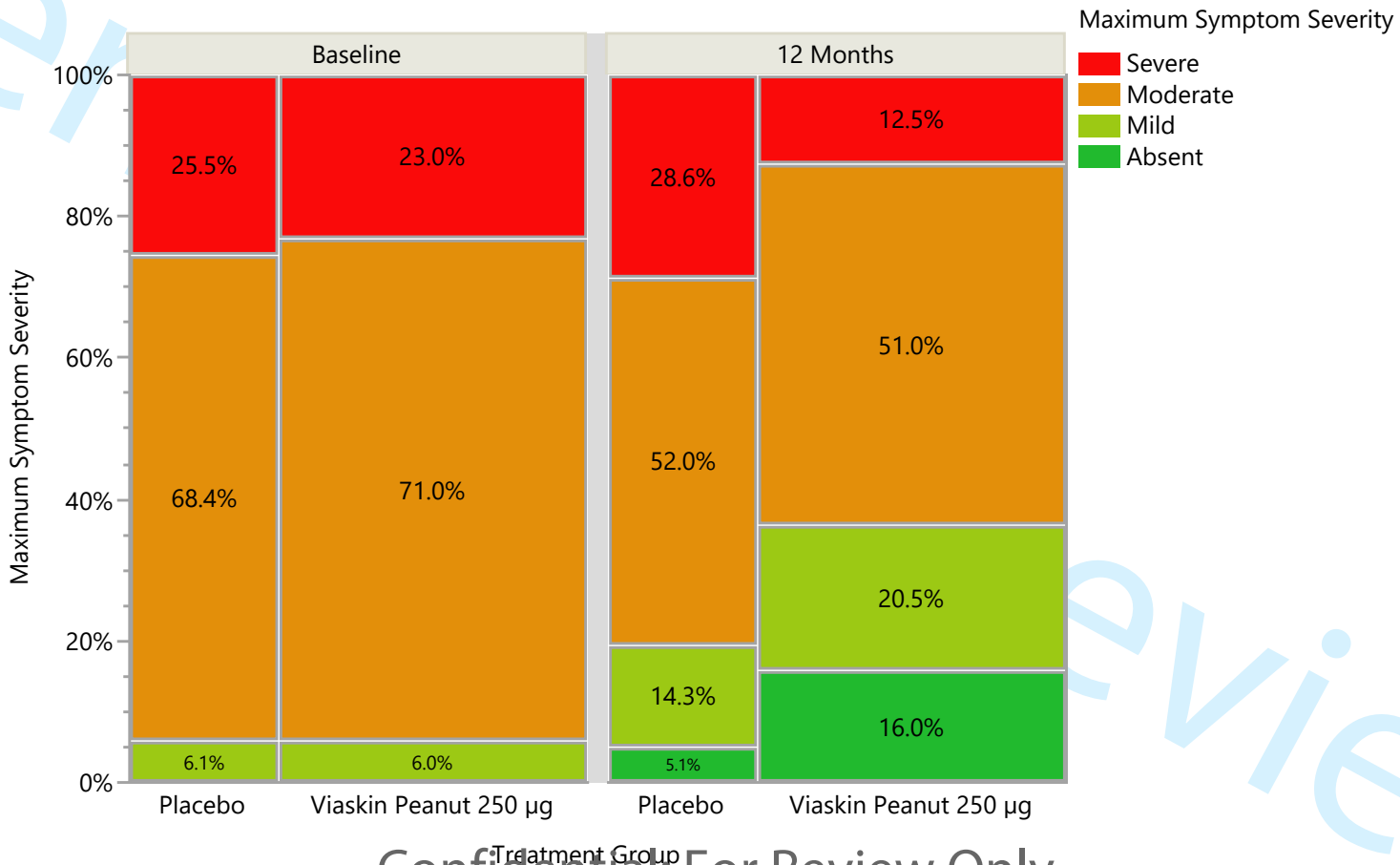

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Figure 1. Participant Disposition CONSORT Diagram.

Eligible participants were randomized 2:1 to receive Viaskin Peanut 250 µg or placebo.

VP250=Viaskin Peanut 250 µg.

Figure 2. Primary (A) and Sensitivity (B) Analyses of the Primary Endpoint.

A. The primary measure of treatment effect compared the percentages of treatment responders between the Viaskin Peanut 250 µg and placebo groups after 12 months of EPIT treatment among all randomized subjects. A subject was defined as a treatment responder if the baseline ED was >10 mg peanut protein and the Month 12 ED was ≥1000 mg peanut protein, or the initial ED was ≤10 mg peanut protein and the Month 12 ED was ≥300 mg peanut protein.

B. Prespecified sensitivity analyses of the primary endpoint were defined as S1: Primary efficacy analysis using missing=failure imputation method for intercurrent events 1, 2, and 5; S2: Primary efficacy analysis using multiple imputation method for intercurrent events 1, 2, and 5; S3: Primary efficacy analysis using multiple imputation method for intercurrent events 2 and 5 (active subjects mimic Viaskin Peanut 250 µg response and placebo subjects mimic placebo response); S4: Primary efficacy analysis excluding subjects with intercurrent event 1, 2, or 5; S5: Primary efficacy analysis excluding patients whose DBPCFC falls outside the Month 12 time-window (intercurrent event 4); S6: Primary efficacy analysis on the per-protocol set.

ED=eliciting dose; S=sensitivity analysis; DBPCFC=double-blind, placebo-controlled food challenge.

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Figure 3. Maximum Symptom Severity at Month 12 Double-Blind Placebo-Controlled Food Challenge.

The severity of symptoms elicited during the DBPCFC was measured at baseline and Month 12 in each treatment group. Severity was assessed by maximum grade of PRACTALL in any PRACTALL symptom domain at each dose.[1] Subjects with intercurrent events 1, 2, 3 and 5 were excluded from this analysis since a Month 12 peanut food challenge was not completed.

Reference

1. Sampson HA, Gerth van Wijk R, Bindslev-Jensen C, et al. Standardizing double-blind, placebo-controlled oral food challenges: American Academy of Allergy, Asthma & Immunology-European Academy of Allergy and Clinical Immunology PRACTALL consensus report. J Allergy Clin Immunol 2012;130(6):1260-74. doi:10.1016/j.jaci.2012.10.017

Figure E1. Study Schema.

EPITOPE was a Phase 3, randomized, double-blind, placebo-controlled study in children 1-3 years of age with confirmed peanut allergy. A total of 362 participants from 51 sites across Australia, Canada, Europe and the US were randomized 2:1 to receive 12 months of EPIT with either Viaskin Peanut 250 µg (N=244) or placebo (N=118). A double-blind, placebo-controlled food challenge was performed at Month 0 (baseline) and Month 12.

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Figure E2. Proportions of Treatment Responders by Baseline Eliciting Dose

The percentage of treatment responders among participants treated with Viaskin Peanut 250 µg (green) or placebo (gray) was analyzed according to baseline eliciting dose (1, 3, 10, 30, 100, or 300 mg). Results displayed in the top graph were calculated without imputation; results displayed in the bottom graph were calculated using missing=failure imputation.

Sum=number of treatment responders; N=total number of subjects; ED=eliciting dose.

Figure E3. Subgroup Analyses of the Primary Endpoint.

Subgroup analyses of the primary endpoint assessed the percentage of treatment responders in Viaskin Peanut µg treated subjects (green bar) and placebo subjects (gray bar) according to eliciting dose (ED) at baseline (≤10 mg or >10 mg), gender, race, and ongoing medical history at baseline. The risk difference with 95% CI between treatment groups and proportions of treatment responders within each treatment group are shown.

Figure E4. Participant-Specific Trajectories in Peanut-sIgE (A) and IgG4 (B) From Baseline to Month 12.

Peanut-sIgE (A) and IgG4 (B) levels were collected at baseline, Month 3, Month 6 and Month 12 of treatment. The box plots extend from the 25th to 75th percentile. The notch around the median displays a non-parametric confidence interval calculated by the median +/- 1.58 x IQR/

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$\sqrt{n}$. The box whiskers extend to the last data point that is within 1.5*IQR from the ends of the box. Individual participant trajectories over time are shown by the faint grey lines. IgE, immunoglobulin E; IgG4, immunoglobulin G4; IQR, interquartile range; kU_A/L, kilounits of antibody per liter.

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